Multitask Learning for Geometric Shape Classification and Counting

Krzysztof Hałubek

1 Exploratory Data Analysis (EDA)

Before training the model, we performed an exploratory analysis to understand the dataset. We examine the distribution of shape counts, the 135 configuration classes, and visualize some sample images.

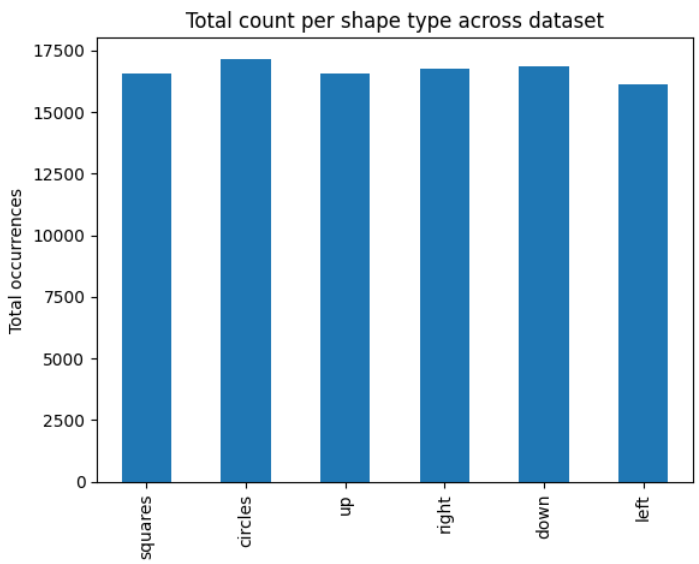


Figure 1: Counts of each shape across the whole dataset.

The dataset contains six types of geometric shapes (squares, circles, and triangles in four orientations). As shown in Figure 1, the shapes are roughly evenly distributed throughout the dataset.

Statistic	Value
Minimum class frequency	0
Maximum class frequency	120
Mean class frequency	74.1
Classes with 0 samples	30

Table 1: Summary statistics of the 135 configuration classes in the dataset.

The dataset covers 135 configuration classes, but their distribution is highly uneven. Class frequencies range from 0 to 120 samples, with a mean of 74.1. Notably, 30 classes contain no samples at all. Such missing classes can make learning more difficult, as the model cannot train on configurations that are entirely absent and may struggle with others that are only weakly represented. Random data augmentations have been employed to combat that.

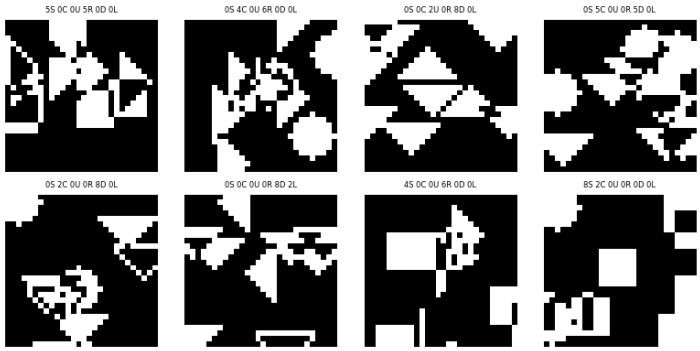


Figure 2: 8 random samples visualized.

Figure 2 shows several randomly selected images from the dataset. Each image contains exactly two types of geometric shapes, with a total of 10 shapes. The shape arrangement can overlap and the images are small (28×28), which can challenge the model's fine-grained distinctions.

2 Model Architecture

The model is a multitask neural network designed for both classification and regression on the Geometric Shape Numbers (GSN) dataset. It consists of a shared convolutional backbone (given as part of the assingment), followed by two separate heads: one for classification (135 classes) and one for regression (6 shape counts).

Classification head (head_cls): predicts the 135 configuration classes using two fully connected layers with ReLU in between and a final LogSoftmax activation.

Layer	Type	Output Shape	Activation
1	Linear(256, 128)	128	ReLU
2	Linear(128, 135)	135	LogSoftmax

Table 2: Classification head layers.

Regression head (head_cnt): predicts the 6 shape counts using two fully connected layers with ReLU in between and a linear output layer.

Layer	Type	Output Shape	Activation
1	Linear(256, 128)	128	ReLU
2	Linear(128, 6)	6	Linear

Table 3: Regression head layers.

3 Data Augmentation

To improve generalization and make the model more robust to variations in the dataset, we applied several data augmentations during training. The following augmentations were implemented:

- **Random horizontal flip:** with 50% probability, the image is flipped along the vertical axis. Triangle orientation counts are updated accordingly to preserve label correctness.
- **Random vertical flip:** with 50% probability, the image is flipped along the horizontal axis, with orientation counts adjusted.
- **Random 90° rotations:** the image is rotated by 0, 90, 180, or 270 degrees with equal probability. Triangle orientations are remapped so that labels remain accurate.

Justification

These augmentations were chosen to reflect natural symmetries in the dataset:

- Flipping and rotation do not change the fundamental shape counts, but expose the model to different spatial arrangements and orientations.
- By applying these transformations, the model is encouraged to learn **rotation- and flip-invariant features**, improving robustness and potentially reducing overfitting.
- Augmentations were applied only to training images to preserve validation set integrity.

4 Results

Below are the validation results of the best models found during the 3 experiments carried out.

Experiment	Loss	Accuracy	RMSE
Classification-only	1.2398	0.486	3.630
Regression-only	0.1183	0.003	0.528
Multitask	2.0181	0.507	0.546

Table 4: Validation results

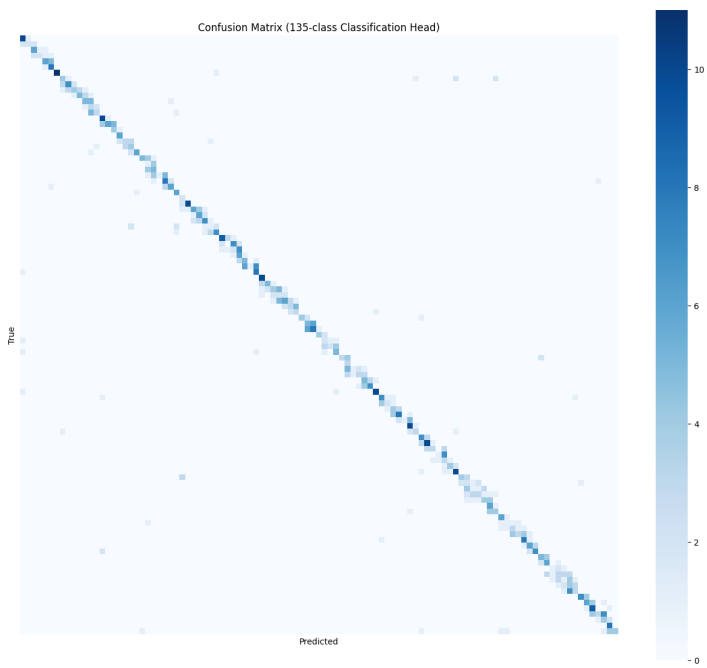
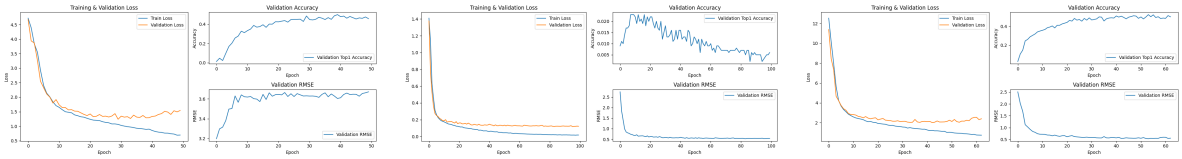


Figure 3: Confusion matrix for Multitask model.

5 Learning Curves



(a) Classification-only (b) Regression-only (c) Multitask

Figure 4: Learning curves for all models.

6 Error Examples

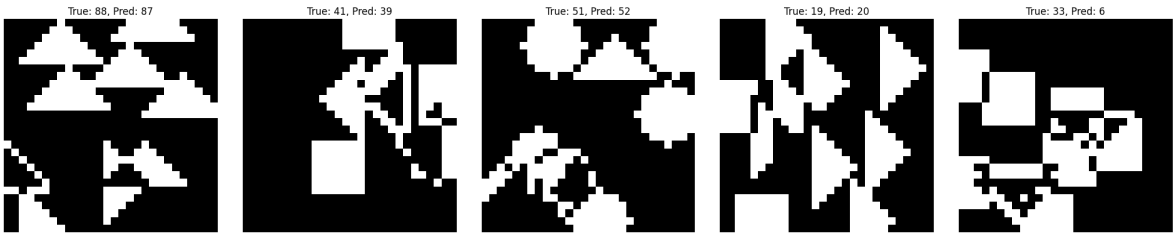


Figure 5: Random Error Examples

The misclassifications reflect mostly small shifts in the counts of the shapes rather than completely wrong shape pairs. For instance, a true class of 88 predicted as 87 or 51 predicted as 52 corresponds to the same pair of shapes but an "off by one" count allocation. Similarly, $41 \rightarrow 39$ and $19 \rightarrow 20$ are also minor count misassignments. The more extreme case, $33 \rightarrow 6$, represents a more serious error where the model misidentified the shape pair itself. Overall, these examples indicate that the model usually predicts the correct pair of shapes but occasionally fails on the precise counts, which is expected with the high difficulty of distinguishing fine-grained count differences in small 28×28 images.

7 Discussion on multitask effects

We trained three models on the GSN dataset: classification-only, regression-only, and a multitask model combining both objectives. For the multitask model, we set $\lambda_{\text{cnt}} = 6$ to roughly balance the magnitudes of the classification and regression losses. The multitask model achieves similar classification performance to the classification-only model (Top-1 Acc 0.507 vs 0.486) and a similar regression performance to the regression-only model (Overall RMSE 0.546 vs 0.528).

The multitask model slightly improves classification compared to the classification-only model, while only modestly increasing regression error relative to the regression-only model. This indicates that the two objectives do not strongly interfere with each other and that their gradients may be partially aligned.. Overall, the multitask approach allows the network to solve both tasks simultaneously with only minor performance trade-offs.